

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                                     |
|----------------------|-------------------------------------|
| United States Patent | 12411250                            |
| Kind Code            | B2                                  |
| Date of Patent       | September 09, 2025                  |
| Inventor(s)          | Rubtsov; Dmitry Anatolyevich et al. |

---

### Method and apparatus for fast searching global navigation satellite system signals

---

#### Abstract

A method and apparatus for fast searching GNSS signals performed on a GNSS receiver includes the steps of receiving a signal having a known pseudo random noise code. State information of a code generator is stored when a pseudo random noise code is generated. Several NCO, including a Doppler NCO are used to search GNSS signal for several supposed Doppler's simultaneously. A search window associated with the received signal is reviewed a first time to identify a source of the received signal. After it is determined if a source of the received signal can be identified, the state information is loaded into the code generator prior to reviewing the search window a second time etc. Search windows is shifting by all length PRN Code. The loading of state information allows sequential review of the search window without re-adjustment of a fast search module which speeds the process of analyzing the received signals.

---

|                              |                                                                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inventors:</b>            | <b>Rubtsov; Dmitry Anatolyevich (Moscow, RU), Purto; Leonid Victorovich (Moscow, RU), Veitsel; Andrey Vladimirovich (Moscow, RU), Prasolov; Victor Andreevich (Moscow, RU)</b> |
| <b>Applicant:</b>            | <b>Topcon Positioning Systems, Inc. (Livermore, CA)</b>                                                                                                                        |
| <b>Family ID:</b>            | <b>85412932</b>                                                                                                                                                                |
| <b>Assignee:</b>             | <b>Topcon Positioning Systems, Inc. (Livermore, CA)</b>                                                                                                                        |
| <b>Appl. No.:</b>            | <b>17/753821</b>                                                                                                                                                               |
| <b>Filed (or PCT Filed):</b> | <b>August 31, 2021</b>                                                                                                                                                         |
| <b>PCT No.:</b>              | <b>PCT/RU2021/000375</b>                                                                                                                                                       |
| <b>PCT Pub. No.:</b>         | <b>WO2023/033667</b>                                                                                                                                                           |
| <b>PCT Pub. Date:</b>        | <b>March 09, 2023</b>                                                                                                                                                          |

## Prior Publication Data

### Document Identifier

US 20230305168 A1

### Publication Date

Sep. 28, 2023

---

## Publication Classification

**Int. Cl.:** **G01S19/30** (20100101); **G01S19/29** (20100101); **G01S19/37** (20100101); G01S19/24 (20100101)

### U.S. Cl.:

**CPC** **G01S19/30** (20130101); **G01S19/29** (20130101); **G01S19/37** (20130101); G01S19/24 (20130101)

## Field of Classification Search

**CPC:** G01S (19/29); G01S (19/30); G01S (19/37); G01S (19/24)

**USPC:** 342/357.68; 342/357.69; 342/357.77; 342/357.63

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name    | U.S. Cl.   | CPC         |
|--------------|-------------|------------------|------------|-------------|
| 6118808      | 12/1999     | Tiemann          | 375/E1.031 | G01S 5/0036 |
| 7395155      | 12/2007     | Hsu              | 370/335    | G01S 19/29  |
| 7630430      | 12/2008     | Bochkovskiy      | 375/150    | H04B 1/707  |
| 7852264      | 12/2009     | Zaruba           | 342/357.69 | G01S 19/37  |
| 8111735      | 12/2011     | Chen             | 375/150    | G01S 19/30  |
| 8170086      | 12/2011     | Abraham          | 375/150    | G01S 19/256 |
| 8193980      | 12/2011     | Nayyar           | N/A        | N/A         |
| 8391339      | 12/2012     | Wang             | 375/150    | H04B 1/7085 |
| 8630332      | 12/2013     | Young            | N/A        | N/A         |
| 8681042      | 12/2013     | Martin           | 342/357.59 | G01S 19/30  |
| 8842717      | 12/2013     | King             | 375/150    | H04B 1/7077 |
| 9596610      | 12/2016     | Borras et al.    | N/A        | N/A         |
| 10281556     | 12/2018     | Mahmood et al.   | N/A        | N/A         |
| 10809385     | 12/2019     | Rubtsov          | N/A        | G01S 19/33  |
| 10880029     | 12/2019     | Lennen           | N/A        | N/A         |
| 10942280     | 12/2020     | Anderson         | N/A        | G01S 19/37  |
| 10943461     | 12/2020     | Kleinbeck et al. | N/A        | N/A         |
| 10945146     | 12/2020     | Kleinbeck et al. | N/A        | N/A         |
| 2002/0193108 | 12/2001     | Robinett         | N/A        | N/A         |
| 2003/0012874 | 12/2002     | Lange et al.     | N/A        | N/A         |
| 2003/0081660 | 12/2002     | King et al.      | N/A        | N/A         |
| 2004/0095275 | 12/2003     | Warloe et al.    | N/A        | N/A         |
| 2008/0232441 | 12/2007     | Mester et al.    | N/A        | N/A         |

|              |         |                  |         |            |
|--------------|---------|------------------|---------|------------|
| 2009/0189808 | 12/2008 | Chen et al.      | N/A     | N/A        |
| 2010/0159958 | 12/2009 | Naguib           | 455/457 | G01S 5/021 |
| 2011/0181467 | 12/2010 | Samavati et al.  | N/A     | N/A        |
| 2012/0038512 | 12/2011 | Geswender et al. | N/A     | N/A        |
| 2013/0035099 | 12/2012 | Boejer et al.    | N/A     | N/A        |
| 2014/0028499 | 12/2013 | Yeh et al.       | N/A     | N/A        |
| 2016/0245923 | 12/2015 | Badke            | N/A     | N/A        |
| 2018/0356531 | 12/2017 | Naveen et al.    | N/A     | N/A        |
| 2021/0055425 | 12/2020 | Guo et al.       | N/A     | N/A        |
| 2021/0226776 | 12/2020 | Falk             | N/A     | N/A        |
| 2022/0276389 | 12/2021 | Yu               | N/A     | G01S 19/21 |
| 2024/0069215 | 12/2023 | Rubtsov          | N/A     | G01S 19/37 |

## FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 1522507    | 12/2003          | CN      | N/A |
| 1543714    | 12/2003          | CN      | N/A |
| 111399004  | 12/2019          | CN      | N/A |
| 112394374  | 12/2020          | CN      | N/A |
| 113141342  | 12/2020          | CN      | N/A |
| 3413089    | 12/2018          | EP      | N/A |
| 2005525005 | 12/2004          | JP      | N/A |
| 2013518281 | 12/2012          | JP      | N/A |

## OTHER PUBLICATIONS

International Search Report and Written Opinion mailed May 19, 2022 in connection with International Application No. PCT/RU2021/000375, filed Aug. 31, 2021, 10 pgs. cited by applicant  
First Office Action mailed Feb. 12, 2025 in connection with Chinese Patent Application No. 202180103206.5, filed Apr. 10, 2024, 13 pgs. (including translation). cited by applicant  
First Office Action mailed Sep. 7, 2024 in connection with Chinese Patent Application No. 202180101963.9, filed Feb. 28, 2024, 12 pgs. (including translation). cited by applicant  
Extended European Search Report mailed May 8, 2025, in connection with European Patent Application No. 21960762.9, filed May 1, 2024, 9 pgs. cited by applicant  
Office Action mailed Jul. 1, 2025, in connection with Japanese Patent Application No. 2024-513488, filed Feb. 28, 2024, 9 pgs. (including translation). cited by applicant

---

*Primary Examiner:* Nguyen; Chuong P

*Attorney, Agent or Firm:* Chiesa Shahinian & Giantomasi PC

---

## Background/Summary

### FIELD OF THE INVENTION

(1) The present disclosure relates to navigation receivers and methods of signal processing and, in particular, to fast searching Global Navigation Satellite Signals (GNSS) and further processing signals of different systems such as the Global Positioning System (GPS), Global Navigation Satellite System (GLONASS), and GALILEO satellite system, etc.

### BACKGROUND

(2) Global Navigation Satellite Systems (GNSS) use satellites to broadcast radio signals that are

acquired by receivers. The receivers use the acquired signals in order to determine the location of the receiver. GNSS signals are often searched for using code delay and Doppler offset. Defining search window  $S$  as the number of simultaneously considered delays viewed by a search unit, the simplest way to search a plurality of channels uses components comprising a code generator, multiple numerically-controlled oscillators (NCO) including a code NCO (CRNCO) and an intermediate frequency NCO (IFNCO), and a correlator. Channels are configured to search one signal for one Doppler offset and, when initializing, a different code delay needs to be set for different channels. To search for a certain Doppler offset, a channel should be re-started/reset. This method requires a great number of channels wherein each is associated with its own correlator, code generator, CRNCO and IFNCO.

## SUMMARY

(3) The present disclosure relates generally to global navigation satellite systems (GNSS) and, more particularly, to a receiver for a GNSS system. In one embodiment, an apparatus for fast searching radio navigational signals having a known pseudo-random noise (PRN) code includes an antenna for receiving signals having a known PRN code. A radio frequency path is configured to receive the radio signals from the antenna and move those signals to an intermediate frequency signal. A numerically controlled oscillator is configured to output pulses at a period of PRN elements and an analog to digital converter (ADC) is configured to sample the intermediate frequency signal. A digital mixer is configured to receive signals from the ADC and output a sampled signal at a zero frequency. A decimator is configured to receive the sampled signal at the zero frequency from the digital mixer. A correlator is configured to calculate a convolution of shifted array of inputs received from the decimator via a pair of quantization units with non-shifted array of PRN code elements and a memory unit is configured to store a result of a value output from the correlator. A code generator is configured to calculate a new element according to the pulses output from the numerically controlled oscillator. An intermediate frequency numerically controlled oscillator is configured to output an intermediate frequency for the intermediate frequency signal. A fast search numerically controlled oscillator (FSNCO) outputting pulses at a preset period. In response to the output pulses of the FSNCO: the decimator is further configured to generate a new output sample, the correlator is further configured to shift a shifted array of input samples to include the new output sample, the correlator is further configured to shift a shifted array of PRN code elements to include a current state of a PRN code generator's output, the correlator is further configured to copy the shifted array of PRN code elements to non-shifted array of PRN code elements one time during  $S$  pulses of the numerically controlled oscillator, the correlator is further configured to calculate a new convolution value and a metric of the new convolution value, a corrector is configured to compare the metric of the new convolution value with the stored result, and, if the new convolution value is greater than the stored result, then the new convolution value is stored instead of the stored result, and a fast search module is configured to determine the availability of a signal with a known PRN code and its parameters in the received radio signal once every  $S \cdot k$  pulses for at least one value of the stored result.

(4) In one embodiment, a Doppler numerically controlled oscillator (DopNCO) is configured to output a Doppler phase once in  $S$  pulses of the FSNCO. In this embodiment, at least  $D-1$  digital phase shifters, where  $D-1$  is an even integer, rotate the new convolution value into a phase proportional to the phase at the DopNCO output. The memory unit is further configured to store  $D \cdot S$  values. In this embodiment, for each pulse of the FSNCO the following operations are performed: in each of  $D-1$  phase shifters, the value output from the correlator is rotated into a phase proportional to the phase of DopNCO output to generate  $D-1$  rotated phase convolution results, obtained  $D-1$  rotated convolution results as well as a non-rotated convolution result are added to previous values in the memory unit configured to store  $D \cdot S$  values, and the obtained  $D$  results of adding are stored in the memory at the same address, and the metric of the new convolution value is calculated according to the result of adding the rotated/non-rotated

convolution result and the previous values in the memory unit configured to store  $D \times S$  values.

(5) In one embodiment of the apparatus, at  $K$ th period of  $S$  pulses, obtained  $D-1$  metrics based on the obtained  $D-1$  rotated convolution results, are input to the memory unit configured to store  $D \times S$  values. Also at  $K$ th period of  $S$  pulses, obtained metrics based on non-rotated results are input to the memory unit configured to store  $D \times S$  values for storing the result, and once every  $S \times k$  pulses, the availability of a signal with a known PRN code and its parameters in the received radio signals is determined.

(6) In one embodiment of the apparatus, a reload generator stores the state of a code generator in  $S+1$  pulse, and, at  $S \times k+1$  pulse when at the end of the incoherent period, the reload generator loads the stored state of the code generator into the code generator.

(7) In one embodiment, the apparatus further comprises a coherent counter and a not-coherent counter, wherein the coherent counter sums convolution values for each  $S$ , and the not-coherent counter is used for  $S$  if needed and the values obtained are stored in the memory unit configured to store  $D \times S$  values.

(8) In one embodiment, the apparatus further comprises a control accumulator comprising  $N$  cyclically shift registers that are moved forward at a rate of the FSNCO, the input to the control accumulator set to 0 based on a configuration.

(9) In one embodiment of the apparatus, addition of estimates during data sorting comprise the results of Doppler metrics during the period  $S \times k$  and Doppler metrics for each offset are separately sorted.

(10) In one embodiment, a method fast searching radio navigational signals includes the step of receiving radio signals at an antenna, the radio signals having a known PRN code. The radio signals from the antenna are transmitted to an RF path which then transmits the signals using an intermediate frequency. The intermediate frequency signal is sampled at an ADC. A digital mixer generates a sampled signal at a zero frequency based on signals received from the ADC. A shifted array of inputs is transmitted from a decimator in response to the decimator receiving the sampled signal at the zero frequency. A convolution of the shifted array inputs received from the decimator via a pair of quantization units is calculated with a non-shifted array of PRN code elements. A result of a value output from the correlator is stored in a first memory unit. A code generator calculates a new element according to pulses output from the numerically controlled oscillator. Pulses at a preset period are output from a fast search numerically controlled oscillator. A decimator generates a new output sample based on the pulses at the preset period. The correlator shifts a shifted array of input samples to include the new output sample. The correlator also shifts a shifted array of PRN code elements to include a current state of a PRN code generator's output. The correlator also copies the shifted array of PRN code elements to non-shifted array of PRN code elements one time during  $S$  pulses of the numerically controlled oscillator. The correlator also calculates a new convolution value and a metric of the new convolution value. A corrector compares the metric of the new convolution value with the stored result, and, if the new convolution value is greater than the stored result, then the new convolution value is stored instead of the stored result. A fast search module determines the availability of a signal with a known PRN code and its parameters in the received radio signal once every  $S \times k$  pulses for at least one value of the stored result.

(11) In one embodiment, a method for fast searching GNSS signals performed on a GNSS receiver includes the steps of receiving a signal having a known pseudo random noise code. State information of a code generator is stored when a pseudo random noise is generated. The pseudo random noise is associated with the pseudo random noise code. A search window associated with the received signal is reviewed a first time to identify a source of the received signal. After it is determined if a source of the received signal can be identified, the state information is loaded into the code generator prior to reviewing the search window a second time. The loading of state information allows sequential review of the search window without re-adjustment of a fast search

module which speeds the process of analyzing the received signals. In one embodiment, the search window is shifting by all length PRN code.

(12) In one embodiment, the received signal is processed by a control accumulator using a plurality of multiplexed signals on a fast search numerically controlled oscillator frequency. One of the plurality of multiplexed signals can be zeroed. This zeroing causes the zeroed signal to be ignored in the analysis of the plurality of signals. In one embodiment, the received signal is multiplied by an intermediate frequency prior to the determining if a source of the signal can be identified. In one embodiment, a rotation angle is added to the received signal. The rotation angle can be based on a Doppler numerically controlled oscillator. The results generated while reviewing the search window can be stored in a coherent mode or a not-coherent mode. The not-coherent mode allows searching for signals with superimposed data.

(13) In one embodiment, at the end of the incoherent period and when the coherent counter counts K periods of S pulses, all not-coherent metric are added, and the result being read by the CPU.

(14) In one embodiment, at the end of the incoherent period and when the coherent counter counts K periods of S pulses, the availability of a signal with a known PRN code and its parameters in the received radio signals is determined among  $D \cdot S$  not-coherent metrics, and the result being read by the CPU,

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a block diagram of a navigation receiver;
- (2) FIG. 2 shows details of the fast search module shown in FIG. 1;
- (3) FIG. 3 shows details of the digital mixer and decimator shown in FIG. 2;
- (4) FIG. 4A shows details of the control counters shown in FIG. 2;
- (5) FIG. 4B shows an example of adjustment of the control counters shown in FIG. 4A and FIG. 2;
- (6) FIG. 4C shows an additional example of adjustment of the control counters shown in FIG. 4A and FIG. 2;
- (7) FIG. 4D shows an additional example of adjustment of the control counters shown in FIG. 4A and FIG. 2;
- (8) FIG. 5 shows details of the partial parallel correlator shown in FIG. 2;
- (9) FIG. 6 shows details of the searchers shown in FIG. 2;
- (10) FIG. 7A shows details of the control accumulation unit shown in FIG. 2;
- (11) FIG. 7B shows standard operation of the control accumulation unit shown in FIG. 7A;
- (12) FIG. 7C shows a non-accumulation mode of operation of the control accumulation unit shown in FIG. 7A;
- (13) FIG. 8 shows details of the corrector shown in FIG. 2;
- (14) FIG. 9 shows a detail of the MAX shown in FIG. 6;
- (15) FIG. 10 shows a flow chart of operation of sorting unit shown in FIG. 9;
- (16) FIG. 11 shows details of the packer shown in FIG. 2;
- (17) FIG. 12 shows signals generated by Doppler NCO shown in FIG. 2; and
- (18) FIG. 13 is a flowchart showing a method for operation of the navigation receiver shown in FIG. 1.

### DETAILED DESCRIPTION

- (19) A method and apparatus for fast searching of satellite signals comprises a receiver receiving and processing signals transmitted from global navigation satellite system satellites.
- (20) FIG. 1 shows receiver **110** for receiving and processing satellite signals. In one embodiment, a satellite signal including pseudo random noise ("PRN") is received by antenna **100**. The received signal passes through RF-path **101(1)** to analog to digital convertor (ADC) **102(1)**. From ADC

**102(1)**, the converted signal is transmitted to satellite channel **103(1)** and Fast Search Module (FSM) **104(1)**. Satellite channel **103(1)** and FSM **104(1)** receive a digitized signal transferred to an intermediate frequency. FSM **104(1)** implements signal searching based on the intermediate frequency and a reference code delay. Satellite channel **103(1)** processes the digitized signal from ADC **102(1)**. It should be noted that multiple sets of RF paths **101(1)** to **101(R)**, ADC **102(1)** to **102(R)**, satellite channel **103(1)** to **103(C)**, and FSM **104(1)** to **104(F)** can be utilized. It should be noted that in cases where multiple similar paths are shown in a figure, only one channel may be described and the other, similar paths, should be understood to be configured and function similarly to the path described.

(21) Timing module **105** synchronizes control of FSM **104(1)** and satellite channel **103**. Timing module **105** counts out the pre-set number of clock pulses and generates interruption in central processing unit (CPU) **106**. CPU **106** controls timing module **105**, FSM **104** and satellite channel **103**. CPU **106** processes the information from FSM **104** and Channel **103** and transmits data to user **108** via communication module **107**.

(22) FIG. 2 shows details of FSM **104(1)** shown in FIG. 1. Although only the configuration and operation of FSM**104(1)** is described herein, additional fast search modules used in receiver **110** are configured and operate similarly. In one embodiment, FSM **104(1)** comprises the following components which interact with various signals. FSM **104(1)** includes a code rate numerically controlled oscillator (NCO) **201** (referred to as a CRNCO), code generator **202**, intermediate frequency NCO (IFNCO) **204**, reference code (reference Pseudo Random Noise (PRN) sequence) **S203**, decimator **205**, Fast Search NCO (FSNCO) **206**, divided fast search frequency **S207**, quantization unit **208**, quantization unit **209**, partial parallel correlator **210**, the number of “units 1” for component I **S211**, the number of “units 1” for component Q **S212**, rotation unit **213**, doppler NCO (DopNCO) **214**, rotated signal (D . . . 2) **S215**, searcher **216**, signals from control counters **S217**, signal of ending the operation of the delay counter (equal to **S403**) **S218**, control counters **219**, packer **220**, memory unit **221**, commutator **222**, control accumulator **223**, reload generator **224**, divider **225**, corrector **226**, correlation signal of component I **S227**, correlation signal of component Q **S228**, signal of ending the operation of the delay counter via the initial unit; (equal to **S417**) **S229**, signal of ending the operation of the not-coherent counter **S230**, digital mixer **231**, control searcher **232**, signal of reading from the memory unit **S233**, signal of writing to the memory unit **S234**, and intermediate frequency signal **S235**.

(23) In one embodiment, CPU **106** controls the following units within FSM **104**: code rate NCO (CRNCO) **201**, code generator **202**, fast search NCO (FSNCO) **206**, quantization unit **208**, quantization unit **209**, partial parallel correlator **210**, Doppler NCO (DopNCO) **214**, searcher **216**, control counters **219**, packer **220**, commutator **222**, control accumulator **223**, reload generator **224**, divider **225**, and corrector **226**.

(24) In one embodiment, FSM **104** requires initialization prior to searching for a signal selected by CPU **106**. In one embodiment, the following operations are performed during initialization. Commutator **222** electrically connects to one of ADC **102(1)** though **102(R)** based on a desired signal to be analyzed. Control accumulation **223** is adjusted as needed. The frequency of the pseudo random noise generator (PRN) in CRNCO **201** is set and divider **225** is adjusted, if needed. Generator code **202** and reload generator **224** are adjusted, if needed. Intermediate frequency **S235** in the oscillator/generator IFNCO **204** is set. The fast search frequency in the oscillator FSNCO **206** is set. The Doppler frequency in the oscillator DopNCO **214** is set. Values for units **402**, **405**, and **408** are adjusted in Control Counter **219**. The settings in quantizers **208** and **209** are adjusted. And corrector **226**, packer **220**, and partial parallel correlator **210** are adjusted.

(25) After initialization FSM **104** operates as follows according to an embodiment. IFNCO **204**, CRNCO **201**, and FSNCO **206** operate based on a signal from timing module **105**. IFNCO **204** generates intermediate frequency signal **S235** which is fed to digital mixer **231**. A fast search frequency from FSNCO **206** is input to control accumulation **223** and divider **225**. Then a signal

from the selected ADC **102** is fed to control accumulation **223** from commutator **222**. If needed, in control accumulator **223** the input signal is set to 0. The signal from the output from control accumulator **223** is input to digital mixer **231**. The fast search frequency is divided by divider **225**, if necessary. Divider **225** then outputs the divided fast search frequency signal **S207** which is input to decimator **205**, partial parallel correlator **210**, and control counters **219**.

(26) In digital mixer **231**, the signal from Control Accumulator **223** and IFNCO **204** are multiplied and input to decimator **205**. Decimator **205** receives the signals from digital mixer **231** and accumulates and stores them with divided fast search frequency **S207**. The stored signals are input to quantizers **208** and **209**. Quantizers **208** and **209** output quantized signals which are input to partial parallel correlator **210**.

(27) Oscillator CRNCO **201** generates a code frequency which is input to reload generator **224** and code generator **202**. Code generator **202** generates a reference code **S203** which is a PRN code. Reload generator **224** is used for re-initialization of code generator **202** when necessary. Reference code signal **S203** is input to partial parallel correlator **210**. In one embodiment, code generator **202** can generate different code types including multiplexed code, BOC code, MBOC code, Memory Code and others. In one embodiment, unit **202** comprises a frequency code divider and a meander generator for generating a code.

(28) Divided fast search frequency signal **S207** is fed to the input of Control Counters **219**. Control Counters **219** generates control signals **S217**, **S218**, **S229** and **S230**. Signal **S217** is fed to Control Searcher **232**, Searcher **216** and Packer **220**. Signal **S218** is input to DopNCO **214**, signal **S229** is input to partial parallel correlator **210**, and signal **S230** is input to CPU **106**.

(29) Signals from control counters **S217** include the following information: delay number **S401**, coherent counter threshold trigger signal **S406**, signal of finding MAX **S412**, signal of ending the operation of not-coherent counter **S413**, searcher's frequency **S420**, signal of starting accumulation process **S421**.

(30) In partial parallel correlator **210**, the signals from output **208**, output **209**, and signal **S203** are used for correlating with divided fast search frequency **S207**. In-time-correlated signals **S211** and **S212** are output from partial parallel correlator **210**.

(31) Signals **S211** and **S212** are input to corrector **226**. In corrector **226**, mathematical operations depending on the correlation time in partial parallel correlator **210** are produced. Signals **S227** and **S228** are output from corrector **226**.

(32) Signal **S218** is input to DopNCO **214**. Based on signal **S218**, DopNCO **214** generates new rotated signals **S215** ( $D \dots 2$ ). Signals **S227**, **S228** and **S215** are input to rotation unit **213**. In rotation unit **213**, signal **S215** is used to rotate signals **S227**, **S228**.

(33) The signals input to searchers **216(D)** and **216(2)** include signals **S217**, **S233**, **S234**, **S227**, **S228** (e.g., output from rotation units **213(D)** and **213(2)**).

(34) Control searcher **232**, generates a signal **S233** when reading from memory unit **221** and generates a signal **S234** when writing to memory unit **221**. Control searcher **232** transmits information from memory unit **221** to Searcher **216(1)**, **216(2)**,  $\dots$  **216(D)** and communicates with memory unit **221** via packer **220**. Control searcher **232** generates signals **S233** and **S234** based on frequency transmitted by searcher **S420**, and control searcher **232** reads and writes to memory unit **221** via packer **220**.

(35) Searcher **216** performs coherent and not-coherent actions using signals **S227** and **S228** (output from rotation unit **213**) for each delay number **S401** and the results are stored in memory unit **221**. Temporary results of calculations are read and written from/to memory unit **221** via packer **220**. Maximal results are also chosen and saved among all the results at the latest interval of coherent and not-coherent storing. The chosen results are metrics. CPU **106** reads the obtained metrics from searcher **216**.

(36) FIG. 7A shows details of control accumulation unit **223** shown in FIG. 2. Control accumulation unit **223** shown in FIG. 7A includes registers **700(1)**, **700(2)**, **700(3)** through **700(N)**,



and switch **701**. In one embodiment, registers **700** are the cyclically shift registers.

(37) In the given example,  $N=2M$ . Before operation, CPU **106** writes values in register **700**. The output signal from FSNCO **206** is fed to unit **223**. Using Fast Search Frequency, the value from **700(1)** is written to **700(2)**, from **700(2)** it is written to **700(3)**, from **700(N-1)** it is further written to **700(N)**, and from **700(N)** it is written to **700(1)**. The output of register **700(N)** is connected to the input of **700(1)** and to the control input of switch **701**. The output of commutator **222** is fed to the input of switch **701**. The output of switch **701** is connected to the input of Digital mixer **231**. When 0 is available at the output of **700(N)**, the signal from the output of unit **222** is fed to the output of **701**. When 1 is available at the output of **700(N)**, value "0" is fed to the output **701**.

(38) FIG. 7B shows a standard operation mode. Processor **106** writes 0 in all registers **700**. A signal from commutator **222** is input to control accumulation unit **223**. In the standard operational mode, a signal from the input to the output transmitted without any change. In this mode, Divider **225** lets the frequency FSNCO **206** pass without its dividing.

(39) FIG. 7C shows a process of operating in a non-accumulation mode. In this embodiment, the number of registers **700** is equal to 4 identified as registers **700(1)**, **700(2)**, **700(3)** and **700(4)**. Processor (CPU) **106** writes the following values into registers **700**: **700(1)=0**, **700(2)=1**, **700(3)=0**, **700(4)=1**. As an example, the operational mode is described with the absence of one chip of FSNCO **206**. A signal from commutator **222** is input to control accumulation unit **223**. The input signal is set to zero using a chip of FSNCO **206**. One chip of FSNCO **206** is equal to zero at the output, the next chip-signal at the output is equal to the input signal and so on. In this mode, divider **225** divides frequency FSNCO **206** into 2.

(40) FIG. 3 shows details of digital mixer **231** and decimator **205**. Digital mixer **231** comprises cosine unit **300**, sine unit **301**, multiplier **302**, and multiplier **303**. Decimator **205** comprises summing unit **304**, summing unit **305**, register **306**, register **307**, buffer **308**, buffer **309**, switch **310**, and switch **311**.

(41) Digital mixer **231** transfers the digitized signal passed **233** to the zero frequency. Frequency IFNCO **204** is input to digital mixer **231**. Frequency IFNCO **204** is input to cosine unit **300** and sine unit **301** of digital mixer **231**. The output of Cosine unit **300** is input to multiplier **302**, where it is multiplied by the output of control accumulation unit **223**. The output of sine unit **301** is input to multiplier **303**, where it is multiplied by the output of control accumulation unit **223**. The outputs of units **302** and **303** are input to decimator **205**.

(42) The output of unit **302** is input to summing unit **304**, where it is added to the output signal of register **306** passing through switch **310**. The output of unit **303** is input to summing unit **305**, where it is added to the output signal of register **307** passing through switch **311**. The output of unit **304** is input to register **306**. The output of unit **305** is input to register **307**.

(43) A sum of results from units **304** and **305** over time are stored in register **306** and **307**. In accordance with signal S207, values from registers **306** and **307** are written in buffers **308** and **309**. According to signal S207, zero from the output of switch **310** is input to summing unit **304**. And according to signal S207, zero from the output of switch **311** is input to summing unit **305**. The output of buffer **308** is fed to the input of quantization unit **208**. The output of buffer **309** is fed to the input of quantization unit **209**. If needed, the output of control accumulation unit **223** can be set to zero, then, the outputs of multipliers **302** and **303** are zero as well.

(44) Returning to FIG. 7B, according to signal S217, the value of registers **306** and **307** are equal to zero. The input signal of control accumulation unit **223** is transmitted to the output of Decimator **205**, and the values are stored in registers **306** and **307**.

(45) Returning to FIG. 7C, according to signal S217, the value of registers **306** and **307** are equal to zero. Then, control accumulation unit **223** sets values of registers **306** and **307** to zero. Due to this, in registers **306** and **307** there is a value of zero for a certain time. When this setting to a value of zero is over, values again are stored in registers **306** and **307**. The process then begins again.

(46) FIG. 4A shows details of control counters **219** shown in FIG. 2. Control counters **219**, in one

embodiment, comprises the following components which interact with various signals including delay counter **400**, delay number **S401**, threshold delay counter **402**, coherent counter **404**, threshold coherent delay unit **405**, coherent counter threshold trigger signal **S406**, not-coherent counter **407**, threshold not-coherent delay **408**, not-coherent counter threshold trigger signal **S409**, AND gate **410**, AND gate **411**, signal searching for MAX/find MAX signal **S412**, signal identifying ending the operation of the not-coherent counter **S413**, signal identifying ending the operation of the coherent counter **S414**, AND gate **415**, signal identifying ending the operation of the delay counter **S417**, start of operation **418**, AND gate **419**, frequency signal from searcher **S420**, and start of accumulation **S421**.

(47) In the process of initializing FSM **104**, CPU **106** starts control counters **219** and assigns threshold delay counter **402**, threshold coherent delay unit **405**, threshold not-coherent delay unit **408**.

(48) After initialization delay counter **400** is set to 0, coherent counter **404** is set to 0, and not-coherent counter **407** is set to 0. Divided fast search frequency signal **S207** is input to control counters **219**. Signal **207** is also input to delay counter **400**, AND gate **415**, and AND gate **419**.

(49) If **S207** is input to delay counter **400**, **1** is added to the current value. Output signal of unit **400** is input to threshold delay counter **402**. The output of unit **402** is connected to input of AND gate **415**. If the value at the input of threshold delay counter **402** is equal to the threshold set by CPU **106**, then, if **S207** is input to unit **415**, the signal of ending the operation of the delay counter **S417** is generated. According to signal **S417**, delay counter **400** takes value 0.

(50) The signal of ending the operation of the delay counter **S417** is input to delay counter **400**, initial unit **416**, start unit **418**, and partial parallel correlator **210**. Delay number signal **S401** is the output of unit **400**. Delay number signal **S401** is input to searcher **216**. The signal of ending the operation of the delay counter **S417** is the same as **S229**.

(51) Initial unit **416** blocks the first pulse of the signal of ending the operation of the delay counter **S417**, in order to keep zero in units **404** and **407**. Such a blocking corresponds to the initial time (see FIGS. 4B/C/D). Signal of ending the operation of the delay counter **S417** passed through the initial **416** is delay counter end signal passed through the initial is **S403**.

(52) Signal **S403** is input to coherent counter **404**, AND gate **416**, and DopNCO unit **214**. Delay counter end signal passed through the initial module **S403** is the same as **S218**.

(53) If **S403** is input to coherent counter **404**, then **1** is added to the current value. The output of unit **404** is input to threshold coherent delay **405**. The output of unit **405** is connected to input of AND gate **416**. If the value at the input of threshold coherent delay **405** is equal to the threshold set by CPU **106**, then, if **S403** is available at the input of unit **416**, the signal of ending the coherent counter operation **S414** is generated. According to signal **S414**, coherent counter **404** takes value 0.

(54) Signal **S414** is input to coherent counter **404**, not-coherent counter **407**, AND gate **411**. Coherent counter threshold trigger signal **S406** is output from unit **405**. Signal **S406** is input to searcher **216** and operation AND unit **410**.

(55) If **S414** is input to not-coherent counter **407**, **1** is added to the current value. The output signal of unit **407** is input to threshold not-coherent delay unit **408**. The output of unit **408** is connected to AND gate **411**. If the value at the input of threshold not-coherent delay unit **408** is equal to the threshold set by CPU **106**, then, if **S414** is input to unit **411**, the signal of ending the operation of the not-coherent counter **S413** is generated. According to **S413**, unit **407** takes value 0.

(56) Signal **S413** is input to not-coherent counter **407**, searcher **216** and CPU **106**. Note that signal **S413** is the same as **S230** which is fed to CPU **106**. Signal **S409** is output from unit **408**. Signal **S409** is input to AND gate **410** and AND gate **411**. Signals **S406** and **S409** are input to AND gate **410**. If **S406** and **S409** are input unit **410**, the signal of searching for MAX **S412** is generated.

(57) Signal **S417** is input to start unit **418**. The output of unit **418** is input to AND gate **419**. The output signal of unit **418** is input to AND gate **419** which does not allow **S207** to pass through AND **419**, until the first pulse of **S417** occurs. That is, the signal output from AND gate **419** occurs after

delay counter **400** has counted up to the threshold assigned in unit **402**, then reset (the full cycle of the delay counter **400** takes place).

(58) Signal **S420** is the output signal of unit **419**. Signal **S421** is fed to the output of start unit **418**. After initialization, **S421**=1. When the first pulse of signal **S417** is input to start unit **418**, signal **S421** is value 1. When the second pulse of signal **S417** comes to unit **418**, signal **S421** is value 0. Signals **S421** and **S420** are input to searcher **216**.

(59) Delay counter **400** counts 0 up to  $S-1$ , where  $S-1$  is the maximal number being programmed in the threshold delay counter **402**.

(60) Signals from Control Counters **S217** include delay number signal **S401**, coherent counter threshold trigger signal **S406**, signal searching for MAX **S412**, signal indicating ending the operation of the not-coherent counter **S413**; signal identifying frequency of searcher **S420**; and signal indicating start of accumulation **S421**.

(61) FIGS. **4B**, **4C**, and **4D** operational diagrams of control counters **219** for the three configuration versions:

(62) FIG. **4B** shows Example 1 in which: the threshold delay counter **402**=3; the threshold coherent delay **405**=1; and the threshold not-coherent delay **408**=1;

(63) FIG. **4C** shows Example 2 in which: the threshold delay counter **402**=3; the threshold coherent delay **405**=0; and the threshold not-coherent delay **408**=2;

(64) FIG. **4D** shows Example 3 in which: the threshold delay counter **402**=3; the threshold coherent delay **405**=2; and the threshold not-coherent delay **408**=0;

(65) FIG. **5** shows details of partial parallel correlator **210** shown in FIG. **2** which comprises code shift register (in scheme C) **500(1)**, **500(2)**, **500(3)**, **500(S)**, component I shift register (in scheme **1**) **501(1)**, **501(2)**, **501(3)**, **501(S)**, reference code shift register (in scheme RC) **502(1)**, **502(2)**, **502(3)**, **502(S)**, component Q shift register (in scheme Q) **503(1)**, **503(2)**, **503(3)**, **503(S)**, multipliers **504(1)**, **504(2)**, **504(3)**, **504(S)**, multipliers, **505(1)**, **505(2)**, **505(3)**, **505(S)**, summing units **506**, summing units **507**, key **508(S)**, key **509(S)**

(66) In one embodiment, initialization of partial parallel correlator **210** is started by keys **508** and **509**.

(67) In one embodiment, partial parallel correlator **210** operates as follows. Reference code signal **S203** is a bit number having values that can be 1 or 0. Quantizer **208** and **209** output the signs of values fed to the input. A bit number with values 0 and 1 is outputted at the output of units **208** and **209**.

(68) Signal **S207** is input to partial parallel correlator **210**. Code shift registers **500(1)**, **500(2)**, **500(3)**, and **500(S)** fix the data if signal **S207** is available. Reference code signal **S203** is input to code shift register **500(1)**. Then the output signal from **500(1)** is input to shift register **500(2)**. The output signal from shift register **500(2)** is input to shift register **500(3)**. The output signal from shift register **500(3)** is input to shift register **500(S)**.

(69) Component I shift registers **501(1)**, **501(2)**, **501(3)**, and **501(S)** and component Q shift register **503(1)**, **503(2)**, **503(3)**, and **503(S)** fix data if **S207** is present. The signal output from unit **208** is input to shift register **501(1)**. Then, the output signal of shift register **501(1)** is fed to the input of shift register **501(2)**. The output signal from the output of shift register **501(2)** is fed to the input of shift register **501(3)**. The output of shift register **501(3)** is input to shift register **501(S)**. The output signal of unit **209** is input to component Q shift register **503(1)**. Then, the output signal of shift register **503(1)** is fed to the input of shift register **503(2)**. The output signal from the output of shift register **503(2)** is fed to the input of shift register **503(3)**. The output of shift register **503(3)** is input to shift register **503(S)**.

(70) Reference code shift register **502** fixes data if **S229 (S417)** is present. The output signal of **500(1)** is fed to the input of **502(1)**. The output signal **500(2)** is fed to the input of **502(2)**. The output signal of **500(3)** is fed to the input of **502(3)**. The output signal of **500(S)** is fed to the input of **502(S)**. Values in unit **502** does not change until next **S229 (S417)** signal is available.

(71) In units **500**, **501** and **503**, the values are moved according to signal **S207**. The output signals from **501(1)** and **502(1)** are input to multiplier **504(1)**. The output signals from **501(2)** and **502(2)** are input to multiplier **504(2)**. Similarly, the output signals from **501(3)** and **502(3)** are input to multiplier **504(3)**. The output signals from **501(S)** and **502(S)** are input to multiplier **504(S)**. Note that the output signal of unit **504** is a bit. Values of units **501** and **502** are multiplied in unit **504**. The output signal from unit **504** is input to unit **506**.

(72) Part of higher digits of unit **504** goes through key **508**. If needed, part of outputs of unit **504** is not input to unit **506**. In one embodiment, whether the outputs of unit **504** are not input to unit **506** depends on the threshold written by CPU **106** in the threshold counter delay **402**.

(73) Output signals from **503(1)** and **502(1)** are input to multiplier **505(1)**. Output signals from **503(2)** and **502(2)** are input to multiplier **505(2)**. Output signals from **503(3)** and **502(3)** are input to multiplier **505(3)**. Output signals from **503(S)** and **502(S)** are input to multiplier **505(S)**. Output signals from each of **505(1)**, **505(2)**, **505(3)** and **505(S)** are each a bit. Values of units **503** and **502** are multiplied in unit **505**. Output signal from unit **505** is fed to unit **507**.

(74) Part of higher digits of unit **505** goes through key **509**. If needed, part of outputs of unit **505** is not input to unit **507**. In one embodiment, whether the outputs of unit **505** are input to unit **507** depends on the threshold written by CPU **106** in the threshold counter delay **402**.

(75) The number of units **1** fed from outputs of unit **504** is outputted at the output of unit **506**. The number of units **1** fed from outputs of unit **505** is outputted at the output of unit **507**. The sum from the output of unit **506** is connected to the number of “units **1**” for component I **S211**. And the sum from the output of unit **507** is connected to the number of “units **1**” for component Q **S212**.

(76) It should be noted that, in one embodiment, code shift register **500** is a shifted array of elements of PRN code. Component I shift register **501** & component Q shift register **502** are a shifted array of input samples. Reference code shift register **502** is a non-shifted array of elements PRN code. A convolution of shifted array **501** and non-shifted array **502** is calculated in partial parallel correlator **210** using units **504** and **506**, the result of this convolution is signal **S211**.

(77) A convolution of shifted array **503** and non-shifted array **502** is calculated in partial parallel correlator **210** using units **505** and **507**, the result of this convolution is signal **S212**.

(78) FIG. **8** shows details of corrector **226** shown in FIG. **2**. Corrector **226** comprises X2 multiplier **800**, X2 multiplier **801**, summing unit **802**, summing unit **803**, and constant **804**. In one embodiment, initialization of corrector **226** occurs as follows. Before operation, CPU **106** sets a value in unit **804** and the maximal value of a constant in unit **804** is equal to **S**.

(79) In one embodiment, operation of corrector **226** occurs as follows. Signal **S211** is input to X2 multiplier **800** (output of unit **506**). In X2 multiplier **800**, the incoming number is multiplied by 2. The output signal from X2 multiplier **800** is input to summing unit **802**. In summing unit **802**, a constant from the output of unit **804** is subtracted from the value from the output of X2 multiplier **800**. The output of summing unit **802** is input to Searcher **216(1)** and Rotation unit **213**. The output of summing unit **802** is signal **S227**.

(80) Signal **S212** is input to X2 multiplier **801** (output of unit **507**). In X2 multiplier **801**, the input value is multiplied by 2. The output of X2 multiplier **801** is input to constant **803**. In constant **803**, from the value of the output of unit **801** is subtracted the constant from the output of unit **804**. The output of unit **803** is input to searcher **216(1)** and Rotation unit **213**. The output of unit **803** is **S228**.

(81) The signal representing the number “1” at the output **506** (**S211**) is the result of the convolution **501** and **502**, from Corrector **226** the mathematically correct convolution number **501** and **502** is sent to the output **S227**.

(82) The signal representing the number “1” at the output **507** (**S212**) is the result of the convolution **503** and **502**, from corrector **226** the mathematically correct convolution number **503** and **502** comes to the output **S228**.

(83) When initializing FSM **104**, the following values are used with the components identified. Threshold counter delay **402** is equal to **S-1** or smaller, key **508** and key **509** are on or off, and

constant **804** is equal to S or smaller.

(84) The expressions for outputs of Corrector **226** are as follows:  $S227 = \text{summing units } (506) * 2 - \text{constant } (804)$ ; and  $S228 = \text{summing units } (507) * 2 - \text{constant } (804)$ .

(85) The following two examples show connections between **S227**, **S228**, **402**, **508**, **509**, and **804** and the tables show values of outputs and constants and contain a description of different scenarios.  
Example 1

(86) Assume  $S=1023$ . The number of processed delays is  $S=1023$ . The threshold counter delay **402**  $= S - 1 = 1023 - 1 = 1022$ . Counter delay **400** counts from 0 up to 1022. Constant **804**  $= S = 1023$ . **508(S)** and **509(S)** are on.

(87) TABLE-US-00001 Out- Output of Formula put 800 Output summing units of multiplied of  
(506) \* 2 - Con- 506 by 2 802 constant stant Description 1023 2046 1023 1023 \* 2 - 1023 1023  
All units equal +1 are fed to the input of unit 506 Positive units 1023. Negative units 0 1 2  
-1021 1 \* 2 - 1023 1023 A unit 1 is fed to the input of unit 506. Value is -1021, as +1 has  
taken one negative unit from the sum. Positive units: 1. Negative unit 1022. 500 1000 -23 500  
\* 2 - 1023 1023 500 units are fed to the input of unit 508. The value is -23, as +500 has taken  
-500 from the sum Positive units: 500 Negative unit 523.

Example 2

(88) Assume  $S=1023-2=1021$ . Reduce the number of the processed delays by 2. The threshold counter delay **402**  $= (S-1)-2 = (1023-1)-2 = 1020$ . Counter delay **400** counts from 0 up to 1020. Constant **804**  $= S = 1021$ . **508(S)** **509(S)** **508(S-1)** **509(S-1)** is disabled. Value **504(S)** **504(S-1)** not connected to input **508**. Value **505(S)** **505(S-1)** not connected to input **509**.

(89) TABLE-US-00002 Out- Output of Formula put 800 Output summing units of multiplied of  
(506) \* 2 - Con- 506 by 2 802 constant stant Description 1021 2042 1021 1021 \* 2 - 1021 1021  
All units equal +1 are fed to the input of unit 506 Positive units 1021. Negative units 0 1 2  
-1019 1 \* 2 - 1021 1021 One unit 1 is fed to the input of unit 506. Value is -1019, as +1 has  
taken one negative unit from the sum. Positive units: 1. Negative unit 1020. 500 1000 -21 500  
\* 2 - 1021 1021 500 units are fed to the input of unit 508. The value is -21, as +500 has taken  
-500 from the sum Positive units: 500 Negative unit 521.

(90) FIG. 6 shows details of searchers **216(1)**, **216(2)**, and **216(D)** as shown in FIG. 2. In one embodiment, searcher **216** comprises components for receiving and outputting various signals including input correlation signal of component I **S601**, input correlation signal of component Q **S602**, summing unit of component I **603**, summing unit of component Q **604**; read signal component I **S605**, read signal component Q **S606**, read signal estimation **S607**, switch **608**, switch **609**, switch **610**, estimation calculation unit **611**, summing unit for estimation **612**, write signal component I **S613**, write signal component Q **s614**, write signal estimation **S615**, switch **616**, switch **617**, switch **618**, output signal of component I **S619**, output signal of component Q **S620**, output signal of estimated **S621**, MAX **622**, and switch **623**. Signals **S619** and **S620** are the coherent metric. Signal **S621** is the not-coherent metric.

(91) In one embodiment, operation of searcher **216** is as follows. When Searcher **216** interacts with memory **221**, reading/writing data is implemented via packer **220**. In searcher **216**, all the operations are performed in series for signal **S401** which is the address for Memory unit **221**: **S605** **S606** **S607** are read from Memory **221**; **603**, **604** units add input and read-out data for components I, Q; **611** performs estimation of the obtained sums for components I, Q; **612** adds input and read-out estimated data; **S613** **S614** **S615** are written to memory **221** component I, Q and estimation; and **622** selects maximal estimated values.

(92) All operations in searcher **216** are performed according to signal **S420**. Control searcher **232** implements the control of data processing in searcher **216**. Control searcher **232** generates the control of writing/reading data from memory **221**. According to signal read from memory **S233** there is reading of data from memory **221**. According to signal write from memory **S234** data are written to Memory **221**.

(93) After the above operations, the reset signal **S421** is set equal to 1. It remains value 1 during the first and second periods of delay counter **400**.

(94) During the first period of operating the delay counter **400** there is no signal **S420**, within this time period units **500**, **501** and **503** are filled according to signal **S207**, and this time period is the initial time (See FIGS. 4B, 4C, and 4D). According to signal **S417** values from component **500** are re-written to component **502**. Further, components **500** **501** and **503** are filled according to signal **S207**, the values in component **502** do not change, until signal **S417** occurs. The result of convolution for units **501** **502** **503** passes through corrector **226** and is input to searcher **216** as signals **S227** and **S228**. Signal **S601** is the same as **S227** and signal **S602** is the same as **S228**.

(95) Each value of signals **S601** and **S602** corresponds to their delay number **S401**. And delay counter **400** operates in cycles, and signals **S601** and **S602** come depending on cyclically-repeated delay number **S401**. Respectively, input values of signals **S601** and **S602** can be sequentially processed for each delay.

(96) During the second period of operating delay counter **400** signal **S421** is 1. According to signal **S420**, signals **S601** and **S602** are sequentially processed. Component I **S605**, component Q **S606**, and estimation **S607** are read from memory for the current delay **S401**. Signal **S605** is fed to unit **608**, signal **S606** is fed to unit **609**, and signal **S607** is fed to **610**. Since signal **S421** is active, zero is fed to the output of units **608** **609** **610**.

(97) The output signal from unit **608** is input to component **603**. The output signal from unit **609** is input to component **604**. The output signal from unit **610** is input to component **612**.

(98) For the current value **S401**, signal **S601** is input to component **603**. For the current value **S401**, signal **S602** is input to component **604**. Values of signal **S601** and output **608** are added in unit **603**. Values of signal **S602** and output of unit **609** are added in unit **604**. Signal **S619** is the output of unit **603**. Signal **S620** is the output of unit **604**.

(99) At **S421**=1, the data read from the memory are zeroed, and a new coherent or, if needed, not-coherent convolution accumulation is started for units **501**, **502** and **503**.

(100) Signals **S619** and **S620** are input to component **611**. The following mathematical operation is performed by component **601**:

$$\text{Output}_{611} = \sqrt{(\text{S619} * \text{S619} + \text{S620} * \text{S620})}$$

(101) The output value from unit **611** is input to component **623**. If **S406**=0, the output of component **623**=0 as well. If **S406**=1, the output signal from component **611** is input to component **623**. The output signal from component **623** is input to component **612**. Output **610** and output **623** are added in unit **612**. Signal **S621** is the output of component **612**.

(102) At **S406**=1, there is not-coherent accumulation of convolution results for components **500** **501** and **503**. Signal **S619** goes to input of component **616**. Signal **S620** goes to input of component **617**. If **S406** is equal to "1", then outputs **616** and **617** are set to "0". If **S406** is equal to 0, then signal **S619** is fed to the output of component **616**, and **S620** is fed to the output of component **617**. Signal **S613** is the output of component **616**. Signal **S614** is the output of component **617**. Signals **S613** and **S614** are written to memory **221** via packer **220**. If **S406**=1 then coherent accumulation is zeroed for convolution **501**, **502** and (if **S406**=1 then coherent accumulation is zeroed for convolution **501** **502** **503**). At the next period of delay counter **400**, signals **S605** and **S606**=0, i.e., components I and Q are equal to 0, and coherent accumulation of convolution results in units **501**, **502**, and **503** starts anew.

(103) Signal **S621** is input to component **618**. If **S412**=1, then the output value of unit **618**=0. If **S412**=0, then **S621** is fed to the output of component **618**. Signal **S615** is the output of component **618**. Signal **S615** is written to memory **221** via packer **220**.

(104) If **S412**=1, then not-coherent accumulation is zeroed for convolution **501**, **502** and **503** (if **S412**=1 then not-coherent accumulation is zeroed for convolution **501** **502** **503**). At the next period of delay counter **400**, signals **S607**=0, i.e., estimation unit is equal to 0, and not-coherent accumulation of convolution results in units **501**, **502**, and **503** starts anew.

(105) Signals **S613**, **S614** and **S615** are written to memory **221** for the current value of delay number **S401**.

(106) For each delay number **S401**, signal **S601** is added to the value from memory **S605** and stored in memory as signal **S613**. Signal **S601** is stored during the whole operational period of coherent counter **404**.

(107) For each delay number **S401**, signal **S602** is added to the value from memory **S606** and stored in memory as signal **S614**. Signal **S602** is stored during the whole operational period of coherent counter **404**.

(108) For each delay number **S401** at the output of unit **611** values are calculated for signals **S619** and **S620**. For each period of coherent counter **404** a sum of the output value from unit **611** and the value from memory **610**, this sum is stored within operation period of not-coherent counter **407**.

(109) For each delay number **S401** at the end of operation of coherent counter **404**, signals **S613** and **S614** being written to Memory **221** are set to 0 according to **S406**.

(110) For each delay number **S401** at the end of operation of not-coherent counter **407**, signal **S615** being written to Memory **221** is set to 0 according to **S412**.

(111) For each delay number **S401**, if **S412** is available at the end of operating not-coherent counter **407**, signals **S619** **S620** **S621**, results of coherent and not-coherent convolution **501**, **502** and **503** are fed to unit **MAX 622**.

(112) FIG. 9 shows the details of **MAX 622** shown in FIG. 6. The following signals are received by **MAX 622**: output of the component I **S619**, output of the component Q **S620**, output of estimated **S621**, delay number **S401**, find **MAX S412**, signal of ending not-coherent counter operation **S413**, searcher frequency **S420**,

(113) **MAX 622** includes components that transmit and receive various signals including control **MAX 900**, data **901(1) . . . 901(M)**, buffer for sorted data **902(1) . . . 902(M)**, summing units **903**, register **904**, signal indicating that searching for **MAX/find MAX** is occurring **S905**, signal indicating that searching for **MAX** has ended **S906**, buffer **907**, signal indicating search result for a single delay, signal of sorting data **S909(1)**, **S909(2)**, **S909(3)**, . . . **S909(M)**, **AND gate 910**, and sorting signal **S911**.

(114) In one embodiment, **MAX 622** operates as follows. **MAX 622** is a sorting device to sort input data during period of component **407**, the sorted-out data being stored and read by **CPU 106** after completing the operation period of component **407**. When initializing, **901**, **902**, **904**, and **907** are all set equal to zero.

(115) In one embodiment, the following signals are input to **MAX 622**: output of the component I **S619**, output of the component Q **S620**, output of the Estimation unit **S621**, delay number **S401**, signal indicating find **MAX S412**, signal indicating ending of operation of the not-coherent counter **S413**, signal identifying searcher frequency **S420**.

(116) Signals **S412** and **S420** are input to component **910**. If **S412** and **S420** are equal to 1 (signal is available), then the output of unit **910=1** (signal is available). The output of unit **910** is the same as signal **S911**. Signal **S911** is input to control **MAX 900**.

(117) Signals **S619**, **S620**, and **S621**, which were obtained within not-coherent counter **407** operation, come together with each delay number **S401** in **MAX 622**. A search result for a single delay **S908** includes signals: **S401**, **S619**, **S620** and **S621**.

(118) The following signals are input to control **MAX 900**: output of the component I **S619**, output of the component Q **S620**, output of the estimated signal **S621**, delay number **S401**, signal of sort **S911**, and signal indicating ending the operation of the not-coherent counter **S413**.

(119) When **S911** is input to control **MAX 900**, signal **S908** is sorted for each signal **S401**. Sorting **S908** is implemented using signal **S621**. Sorting lasts the whole period of operating not-coherent counter **407**. Signal **S909** is the sorted-out values **S908**. Signals **S909** from unit **900** are input to unit **901(1)**, **901(2)**, **901(3)**, . . . , **901(M)** and stored there. The number of sorted values is **M**. When sorting **S908** is over, value in unit **901** is updated if needed.

(120) When signal **S413** confirming the end of operating not-coherent counter **407** is detected, control **MAX 900** waits for sorting end signal **S908** for the last value of **S401**. When sorting is completed, signal **S906** is output. Sorted values **S909** are written to data **901(1)**, **901(2)**, **901(3)** . . . **901(M)**.

(121) Signal **S906** from control **MAX 900** is input to the following components: buffer for sorted data **902(1)**, **902(2)**, **902(3)**, . . . , **902(M)**, register **904**, and buffer **907**. When sorting is completed, the value from unit **901** is written to buffer **902(1)**, **902(2)**, **902(3)**, . . . **902(M)** according to signal **S906**. According to signal **S906**, values in units **901** are set to 0.

(122) Sorting continues for the next operation period of not-coherent counter **407**. Data **902** is stored until next **S906** signal is received. CPU **106** reads values from **902(1)**, **902(2)**, **902(3)**, . . . **902(M)**.

(123) Signal **S905** at the output of unit **900** is the same as signal **S911**.

(124) Signal **S621** is input to adder **903**. The output of register **904** is input to summing units **903** and Buffer **907**. Signal **S621** and the output signal of unit **904** are added in unit **903**. The output of unit **903** is fed to the input of unit **904**. A sum of **S621** values for each delay number **S401** is stored in unit **904** during operation period of not-coherent counter **407** if **S905** (**S911**) is available.

(125) According to signal **S906**, the stored sum **S621** from register **904** is written to unit **907**, and values in the register **904** are set to 0. When the next operation period of not-coherent counter **407** starts, and first signal **S911** arises, the input value of unit **904** is set to 0. A sum of not-coherent values for each signal **S401** is stored in register **904** during operation of signal **S412=1** if signal **S420** is available. The value from Buffer **907** is read by CPU **106**.

(126) FIG. **10** shows a flow chart of a method for sorting search results for a single delay in control **MAX 900** where the method begins at start operation **1001**. A sorting signal is generated at generation signal sort **1002**, and conditions **1003(1)**, **1003(2)**, **1003(3)**, . . . **1003(M)** are checked and operations **1004(1)**, **1004(2)**, **1004(3)**, . . . **1004(M)** are performed based on the conditions.

(127) In one embodiment, the operation of sorting unit in control **MAX 900** is in accordance with the method of FIG. **10** as follows.

(128) Data sorting in control **MAX 900** starts if signal **S911** is detected. Data sorting is performed based on the output signal of the estimation unit **S621**. Signal **S621** is the input estimate. A search result for a single delay **S908** includes signals: **S401**, **S619**, **S620** and **S621**. Estimate 1 is signal **S621** from Data **901(1)**. Estimate 2 is **S621** from Data **901(2)**, and Estimate 3 is **S621** from Data **901(3)**. So, Estimate **M** is signal **S621** from Data **901(M)**.

(129) During initialization unit **901** is set to 0. Once FSM **104** has been initialized, the signal is input to begin at start **1001**. From step **1001** the flow chart goes to step **1002**. At step **1002**, the method analyzes signal **S911**. If **S911=0**, the flow chart loops to step **1001** and then it comes back to unit **1002**. If **S911=1**, the flow chart proceeds to **1003(1)**.

(130) At step **1003(1)**, condition “input estimate is greater than estimate 1” is analyzed. If this condition is not satisfied, then the method proceeds to step **1003(2)**. If condition in step **1003(1)** is satisfied, the following operations are carried out: Data **901(M-1)** is written to Data **901(M)**; Data **901(2)** is written to Data **901(3)**; Data **901(1)** is written to Data **901(2)**; **S908** is written to Data **901(1)**; and then the method returns to start **1001**.

(131) At step **1003(2)** condition “Input estimate is greater than estimate 2” is analyzed. If condition **1003(2)** is not satisfied, the method proceeds to step **1003(3)**. If it is satisfied, the following operations are carried out: Data **901(M-1)** is written to Data **901(M)**; Data **901(2)** is written to Data **901(3)**; **S908** is written to Data **901(2)**; Data **901(1)** does not change; and then the method returns to start **1001**.

(132) At step **1003(3)** condition “Input estimate is greater than estimate 3” is analyzed. If condition **1003(3)** is not satisfied, the method proceeds to step **1003(4)**. If it is satisfied, the following operations are carried out: Data **901(M-1)** is written to Data **901(M)**; **S908** is written to Data **901(3)**; Data **901(2)** does not change; Data **901(1)** does not change; and then the method returns to



start **1001**.

(133) At step **1003(M)** condition “Input estimate is greater than estimate M” is analyzed. If condition **1003(M)** is not satisfied, the method proceeds to start **1001**. If it is satisfied, the following operations are carried out: **S908** is written to Data **901(M)**; Data **901(3)** does not change; Data **901(2)** does not change; Data **901(1)** does not change; and then the method returns to start **1001**.

(134) FIG. **11** shows details of packer **220** shown in FIG. **2**. Packer **220**, in one embodiment, includes components that transmit and receive various signals including data unpacker **1100**, address **S1101**, read data **S1102**, read **S1103**, write **S1104**, write data **S1105**, and pack data **1106**.

(135) In one embodiment, packer **220** operates as described as follows in conjunction with FIGS. **11** and **2**. In the process of signal searching in conjunction with DopNCO **214**, rotation units **213(2)** . . . **213(D)** and searchers **216(1)**, **216(2)**, . . . **216(D)** obtained results are written and read to/from memory **221** via packer **220**. Using signal **S401** as address **S1101**, signals **S613**, **S614** and **S615** pass through data packer **1106** and are written to memory **221** in one word. Write data **S1105** comprises data that is written to memory unit **221** in a similar manner.

(136) Using signal **S401** as address **S1101**, data is read from memory **221** (or similarly read signal read data **S1102**) as one word. Signal **S1102** passing through data unpacker **1100** is unpacked to generate signals **S605**, **S606** and **S607** which are then transmitted to searcher **216**.

(137) The results of operation of searcher **216** are signals **S613**, **S614** and **S615** which are written to memory **221** at address **S1101**. The bit number in signals **S613**, **S614** and **S615** can exceed bit number in the word write data **S1105**. As a result, signals **S613**, **S614** and **S615** can be packed in a different way.

(138) For example, at the input of packer **220** there are integers, but in packer **220** they are transformed into floating format, where the floating number is mantissa and exponent, all the numbers having a common exponent. The mantissa is kept/different for each number, but the exponent is common. The exponent is chosen such that all high-order digits would lie within the mantissa.

(139) Signals **S615(1)**, **S615(2)**, . . . **S615(D)** from outputs of searchers **216(1)**, **216(2)**, . . . **216(D)** are combined by the common exponent. Signals **S613** and **S614** from outputs of searcher **216(1)**, **216(2)**, . . . **216(D)** are also combined by the common exponent. The mantissa and exponents of signals **S615**, **S613** and **S614** obtained at data packing are fed to the output of data packer **1106**. In data pack **1100** data, read from the memory, are unpacked considering mantises, exponents and operation mode of packer **220**.

(140) In one embodiment, packer **220** is initialized when CPU **106** starts operation of packer **220**. In one embodiment, packer **220** operates as follows.

(141) **S401** is input to packer **220**. Signal **S1101** is the same as signal **S401**. Signal **S1101** is input to memory unit **221** as an address. Signal **S1102** is input to data unpack **1100** of data read from memory **221**. In memory unit **221**, **S1105** is input as written data from unit **1106**.

(142) Signal **S233(1)** is input to packer **220**. Signal read **S1103** is the same as signal **S233**. **S1103** is input to memory unit **221** as a read. If signal **S233** is detected, data are read from memory unit **221** at address identified by signal **S1101**, and signal **S1102** goes to the input of unit **1100**, where they are unpacked. From data unpack **1100** these signals are input to: searcher **216(1)** which receives signals **S605(1)**, **S606(1)** and **S607(1)**; searcher **216(2)** which receives signals **S605(2)**, **S606(2)** and **S607(2)**; and searcher **216(D)** which receives signals **S605(D)**, **S606(D)** and **S607(D)**.

(143) Signal **S234(1)** is input to packer **220**. Signal write **S1104** is the same as signal **S234**. **S1104** is input to memory **221** as a write.

(144) In data pack **1106**, signals input are packed as follows: signals **S613(1)**, **S614(1)** and **S615(1)** are output from searcher **216(1)**; signals **S613(2)**, **S614(2)** and **S615(2)** are output from searcher **216(2)**; and signals **S613(D)**, **S614(D)** and **S615(D)** are output from searcher **216(D)**.

(145) If signal **S234** is available, data packed in **1106** are written (write data **S1105**) to Memory

**221** at address **S1101** write data **S1105**.

(146) There are different packing and unpacking modes for storing temporary data in unit **221** depending on searcher **216** modes and the number of Doppler NCO.

(147) Here are Some Examples:

(148) 1) Condition: Unit **405**=0, Unit **408**=0. Result: In this case there no need to keep data from signals **S613 S614 S615** in memory **221**. 2) Condition: Unit **405** is greater than 0, Unit **408**=0. Result: In this case there is no need to keep data from signal **S615** in memory **221**. In memory **221** there are kept only data from signals **S613** and **S614**. 3) Condition: The capacity of signal **S613** and signal **S614** has a greater priority than capacity of signal **S615**. Result: To pack data from signal **S613** and signal **S614**, a lower-digit exponent and higher-digit mantissa are used. For signal **S615**, a higher-digit exponent and lower-digit mantissa are kept. 4) Condition: The capacity of signal **S615** has a greater priority than capacity of signal **S613** and signal **S614**. Result: To pack data from signal **S615**, a lower-digit exponent and higher-digit mantissa are used. For signal **S613** and signal **S614**, a higher-digit exponent and lower-digit mantissa are kept. 5) Condition: In memory **221** there is enough space for values of signals **S615**, **S613** and **S614**. Result: Signals **S615**, **S613** and **S614** are written to memory **221**.

(149) FIG. **12** shows details of Doppler NCO **214** shown in FIG. **2**. Doppler NCO **214** generates rotation signal **S215(D)**, . . . **S215(3)**, **S235**, **S215(2)** . . . **S215(D-1)** Each rotation signal **S215** is used to generate Left Doppler or Right Doppler frequency.

(150) Intermediate Frequency **S235** serves as Center Doppler frequency. Right Doppler frequency 1 generates **S215(2)**, Left Doppler frequency 1 is generated mirror-like using signal **S215(3)**. Right Doppler frequency **D/2** generates signal **S215(D-1)** and Left Doppler frequency **D/2** is generated mirror-like using **S215 (D-1)**.

(151) In FSM **104** Doppler Frequency is generated with a step of **F<sub>dop</sub>**, these frequencies include: Center Doppler; Left Doppler **1**; Right Doppler **1**; Left Doppler **D/2**; and Right Doppler **D/2**.

(152) Left Doppler frequency (**D/2 . . . 1**) and Right Doppler frequency (**D/2 . . . 1**) change in phase available by **S218** (equal to **S403**) considering **S** length of shift registers **500, 501, 502, 503** and unit **402** set by CPU **106**. The expression for calculation of **F<sub>dop</sub>** is as follows.

$$F_{dop} = (F_{sub} \cdot IF / S) \cdot d[N-1:0] / 2 \cdot \sup.N$$
 where: **F<sub>sub</sub>.IF** is Intermediate Frequency **S235**; **S** is the length of shift registers **500, 501, 502, 503** (delay counter **400** counts from 0 up to **S-1**); and **N** is the bit number of Doppler NCO phase; **d[N-1:0]** the number written to the Doppler NCO **214**.

(153) Signal **S215** from the output of Doppler NCO **214** is input to rotation unit **213**. Signals of component **I S227** and signal of component **Q S228** are input to rotation unit **213**. In each rotation unit **213** signals **S227** and **S228** are rotated in phase in an angle set by **S215**. Its own rotation signal **S215(D . . . 2)** is generated for Left Doppler (**D/2 . . . 1**) and Right Doppler (**D/2 . . . 1**), **S227** and **S228** rotate phase in a different way.

(154) In unit **213(D)** signals **S227** and **S228** are rotated in phase at frequency Left Doppler **D/2** (**S215(D)** is used) and are fed to the input of Searcher **216(D)**.

(155) In unit **213 (D-1)**, signals **S227** and **S228** are rotated in phase at frequency Right Doppler **D/2** (**S215(D-1)** is used) and are fed to the input of Searcher **216 (D-1)**.

(156) In unit **213(3)**, signals **S227** and **S228** are rotated in phase at frequency Left Doppler **1** (**S215(3)** is used) and are fed to the input of Searcher **216(3)**.

(157) In unit **213(2)**, signals **S227** and **S228** are rotated in phase at frequency Right Doppler **1** (**S215(2)** is used) and are fed to the input of Searcher **216(2)**.

(158) Signals **S227** and **S228** are fed to the input of Searcher **216(1)** at Intermediate Frequency **S235**.

(159) The operation of shift registers **500, 501, 502**, and **503** and reload generator **224** will now be explained using an example including several tables showing values at different points in time. Assume, **S=4**, code length=12, generated frequency CRNCO **201** is equal to frequency of FSNCO **206**. The state of code generator **202** is memorized as 5.<sup>sup</sup>.th chip of FSNCO **206** and uploaded in



to increase search time in  $k$  times. Search time can be increased thanks to coherent accumulation.

An increase in accumulation time can be also possible when information symbols imposed/overlapped on code by not-coherent accumulation are available.

(174) If code length is greater than  $S$ , reload generator **234** can be also used and, after starting operation of FSM **104** the sliding window/search window  $S$  is moved along the whole code length, and metrics can be obtained for all delays.

(175) With Fast Search NCO **206**, different numbers of delays can be set for one CODE chip. For example, two semi-chips are used for one GPS CA chip, and frequency Fast Search NCO **206** is twice as much as CRNCO **201**.

(176) Multiplexed code is used at least as two chip by chip codes. Using fast search NCO **206**, control accumulation unit **223** and divider **225**, one code can be sent into searching mechanism, and the other code set to zero in the input signal. Doppler NCO **214** and Rotation unit **213** allow obtaining results for a plurality of Doppler frequencies for the searching window  $S$ .

(177) Different operational modes of Packer **220** make it possible to select a preferable value: component  $I$  or component  $Q$ , or estimate during data packing/unpacking. At the end of searching period  $S \cdot k$ , when metrics are sorted, estimates of all  $S$  are added.

(178) Using FSM **104** it is possible to receive CSK-modulated signals. For example, in the process of receiving GPS CA signal using semi-chips CODE the frequency of unit Fast Search NCO **206** is twice as much as frequency CRNCO **201**, data are coherently accumulated within 2 milliseconds and  $S=1023$ , threshold not-coherent **408** set **0**. In this case, the convolution results of all registers **501**, **502**, and **503** are coherently stored during 4 periods of operating delay counter **400** for each  $S$ , at the 4th period unit MAX **622** sorts metrics.

(179) It should be noted that reload generator **224** remembers the state of code generator **202** at the  $1023+1$ -th chip of fast search NCO **206** and records the memorized state of unit **202** so that code generator can start at once from this state in/to the  $1023 \cdot 4 + 1$ -th chip of fast search NCO **206**. Setting data make it possible to view **2046** GPS CA semi-chip CODE without re-adjusting FSM **104**.

(180) FIG. **13** depicts a flow chart of method **1300** for fast searching global navigation satellite system signals according to an embodiment. In one embodiment, method **1300** is performed using navigation receiver **110** shown in FIG. **1**. At step **1302**, a signal is received at the receiver. The signal has a known pseudo random noise code. At step **1304**, state information of a code generator is stored when a pseudo random noise is generated by the code generator. The state information, in one embodiment, is the information pertaining to the code generator that is required to be input into the code generator in order to generate the same pseudo random noise previously generated. In one embodiment, the pseudo random noise generated by the code generator is associated with the pseudo random noise code of the signal. At step **1306**, a search window associated with the received signal is reviewed a first time. The search window, in one embodiment, comprises a string of data obtained from the received signal. As described above, the reviewing is to attempt to identify a source of the received signal (i.e., the satellite that transmitted the received signal). At step **1308**, it is determined if a source of the received signal can be identified. After a source is identified, the information can be used to determine the position of the receiver. At step **1310**, the state information previously stored in step **1304** is loaded into the code generator prior to reviewing the search window a second time in response to the determining. The loading of the state information into the code generator allows sequential review of the search window without re-adjustment of FSM **104**. In one embodiment, the received signal is processed by a control accumulator using a plurality of multiplexed signals on a fast search numerically controlled oscillator frequency. One of the plurality of multiplexed signals can be zeroed. Zeroing one of the signals causes the zeroed signal to be ignored in the analysis of the plurality of signals. In one embodiment, the received signal is multiplied by an intermediate frequency prior to the determining in step **1308**. In one embodiment, a rotation angle is added to the received signal. The

rotation angle can be based on a Doppler numerically controlled oscillator. The results generated while reviewing the search window can be stored in a coherent mode or a not-coherent mode. The not-coherent mode allows searching for signals with superimposed data.

(181) The foregoing Detailed Description is to be understood as being in every respect illustrative and exemplary, but not restrictive, and the scope of the inventive concept disclosed herein is not to be determined from the Detailed Description, but rather from the claims as interpreted according to the full breadth permitted by the patent laws. It is to be understood that the embodiments shown and described herein are only illustrative of the principles of the inventive concept and that various modifications may be implemented by those skilled in the art without departing from the scope and spirit of the inventive concept. Those skilled in the art could implement various other feature combinations without departing from the scope and spirit of the inventive concept.

## Claims

1. An apparatus for fast searching radio navigational signals having known pseudo-random noise (PRN) code, the apparatus comprising: an antenna configured to receive radio signals having a known PRN code; a radio frequency (RF) path configured to receive the radio signals from the antenna and move the radio signals to an intermediate frequency signal; a numerically controlled oscillator configured to output pulses at a period of PRN elements; an analog to digital converter (ADC) configured to sample the intermediate frequency signal; a digital mixer configured to receive signals from the ADC and output a sampled signal at a zero frequency; a decimator configured to receive the sampled signal at the zero frequency from the digital mixer; a correlator configured to calculate a convolution of shifted array of inputs received from the decimator via a pair of quantization units with non-shifted array of PRN code elements; a memory unit configured to store a result of a value output from the correlator; a code generator configured to calculate a new element upon pulses output from the numerically controlled oscillator; an intermediate frequency numerically controlled oscillator configured to output an intermediate frequency for the intermediate frequency signal; and a fast search numerically controlled oscillator (FSNCO) outputting pulses at a preset period, wherein in response to the output pulses of the FSNCO: the decimator is further configured to generate a new output sample; the correlator is further configured to shift a shifted array of input samples to include the new output sample; the correlator is further configured to shift a shifted array of PRN code elements to include a current state of a PRN code generator's output; the correlator is further configured to copy the shifted array of PRN code elements to non-shifted array of PRN code elements one time during S pulses of the fast search numerically controlled oscillator; the correlator is further configured to calculate a new convolution value and a coherent metric of the new convolution value; a corrector is configured to compare the coherent metric of the new convolution value with the stored result, and, if the new convolution value is greater than the stored result, then the new convolution value is stored instead of the stored result; and a fast search module is configured to determine an availability of a signal with a known PRN code and its parameters in the received radio signal once every  $S \cdot k$  pulses of the fast search numerically controlled oscillator for at least one value of the stored result, wherein  $S \cdot k$  is a searching period, S of the searching period is a searching window, and k of the searching period is a multiplier.

2. The apparatus of claim 1, further comprising: a Doppler numerically controlled oscillator (DopNCO) configured to output a Doppler phase once in S pulses of the FSNCO; at least D-1 digital phase shifters, where D-1 is an even integer, rotating the new convolution value into a phase proportional to the phase at the DopNCO output; the memory unit further configured to store  $D \cdot S$  values, wherein for each pulse of the FSNCO the following operations are performed: in each of D-1 phase shifters, the value output from the correlator is rotated into a phase proportional to the phase of DopNCO output to generate D-1 rotated phase convolution results; obtained D-1

rotated convolution results as well as a non-rotated convolution result are added to previous values in the memory unit configured to store  $D \cdot S$  values, and the obtained  $D$  results of adding are stored in the memory at the same address; and the coherent metrics of the new  $D$  convolution values are calculated according to the result of adding the rotated/non-rotated convolution result and the previous value in the memory unit configured to store  $D \cdot S$  values.

3. The apparatus of claim 2, wherein, at  $K$ th period of  $S$  pulses, the availability of a signal with a known PRN code and its parameters in the received radio signals is determined among  $D \cdot S$  coherent metrics.

4. The apparatus of claim 1, further comprising: a reload generator, the reload generator storing the state of code generator, and at  $S \cdot k + 1$  pulse when at the end of the incoherent period, the reload generator loads the stored state of the code generator into the code generator.

5. The apparatus of claim 1, further comprising: a coherent counter; and a not-coherent counter, wherein the coherent counter counts  $K$  periods of  $S$  pulses, and the not-coherent counter can be used to count a period of the coherent counter, wherein at the last  $S$  pulses before the not-coherent counter is incremented, each  $D$  coherent metrics are converted to values to be added to  $D$  not-coherent metrics and the updated not-coherent metrics and the values obtained are stored in the memory unit.

6. The apparatus of claim 5, wherein, when at the end of the incoherent period and when the coherent counter counts  $K$  periods of  $S$  pulses, a fast search module detects/sort several max values in not-coherent metrics  $S \cdot D$  for each Doppler frequency separately, and saves them, saved values read by a CPU, the saved values include not-coherent metrics, coherent metrics, and delay number.

7. The apparatus of claim 1, further comprising: a control accumulator comprising  $N$  cyclically shift registers that are moved forward at a rate of the FSNCO, the input to the control accumulator set to 0 based on a configuration.

8. The apparatus of claim 1, wherein addition of estimates during data sorting comprises the results of Doppler metrics during the period  $S \cdot k$  and Doppler metrics for each offset are separately sorted.

9. The apparatus of claim 1, wherein at the end of the incoherent period and when the coherent counter counts  $K$  periods of  $S$  pulses, all not-coherent metric are added, and the result being read by a CPU.

10. A method comprising: receiving radio signals at an antenna, the radio signals having a known PRN code; transmitting the radio signals from the antenna to an RF path; transmitting the signals from the RF path using an intermediate frequency signal; sampling the intermediate frequency signal at an ADC, the sampling at a period of pulses output from a numerically controlled oscillator of PRN elements; generating, by a digital mixer, a sampled signal at a zero frequency based on signals received from the ADC; transmitting outputs from a decimator in response to the decimator receiving the sampled signal at the zero frequency to a shifted array of inputs; calculating, at a correlator, a convolution of the shifted array inputs received from the decimator via a pair of quantization units with a non-shifted array of PRN code elements; storing a result of a value output from the correlator in a first memory unit; calculating, at a code generator, a new element according to pulses output from the numerically controlled oscillator; outputting, from a fast search numerically controlled oscillator (FSNCO), pulses at a preset period; generating, by the decimator, a new output sample based on the pulses at the preset period; shifting, by the correlator, a shifted array of input samples to include the new output sample; shifting, by the correlator, a shifted array of PRN code elements to include a current state of a PRN code generator's output copying, by the correlator, the shifted array of PRN code elements to non-shifted array of PRN code elements one time during  $S$  pulses of the numerically controlled oscillator; calculating, by the correlator, a new convolution value and a coherent/not-coherent metric of the new convolution value; comparing, by a corrector, a metric based on calculated not-coherent metric value with a stored result, and, if the new metric value is greater than the stored result, then the new convolution value is stored instead of the stored result; and determining, by a fast search module, an availability of a signal with a

known PRN code and its parameters in the received radio signal once every  $S \cdot k$  pulses of Fast Search NCO for at least one value of the stored result, wherein  $S \cdot k$  is a searching period,  $S$  of the searching period is a searching window, and  $k$  of the searching period is a multiplier.

11. The method of claim 10, further comprising: outputting a Doppler phase once in  $S$  pulses of the FSNCO from a Doppler numerically controlled oscillator (DopNCO); rotating, by at least  $D-1$  digital phase shifters, the new convolution value into a phase proportional to the phase at the DopNCO output; storing  $D \cdot S$  values in a the memory unit; rotating, in each of  $D-1$  phase shifters, for each pulse of the FSNCO, the value output from the correlator into a phase proportional to the phase of Doppler NCO output to generate  $D-1$  rotated phase convolution results; adding, in the memory unit, for each pulse of the FSNCO, the obtained  $D-1$  rotated convolution results as well as a non-rotated convolution result to previous values in the memory unit, and the obtained  $D$  results of adding are stored in the memory unit at the same address; and calculating, for each pulse of the FSNCO, the coherent metrics of the new results.

12. The method of claim 11, wherein, obtained  $D-1$  metrics based on the obtained  $D-1$  rotated convolution results, are input to the memory unit configured to store  $D \cdot S$  values; obtained metrics based on non-rotated/rotated results are input to the memory unit configured to store  $D \cdot S$  values for storing the result; and once every  $S \cdot k$  pulses, the availability of a signal with a known PRN code and its parameters in the received radio signals is determined.

13. The method of claim 10, further comprising: storing, at a reload generator, the state of code generator; and loading, at  $S \cdot k + 1$  pulse when at the end of the incoherent period, the stored state of the code generator from reload generator into the code generator.

14. The method of claim 10, further comprising: counting  $K$  periods of  $S$  pulses, the  $K$  periods being associated as coherent integration period; counting a number of coherent integration periods, the number of coherent integration periods being associated as not-coherent integration period; and converting  $D$  coherent metrics to values to be added to  $D$  not-coherent metrics, each converting being performed upon each pulse from the last  $S$  pulses before end of each coherent integration period, the updated not-coherent metrics being stored in the memory unit configured to store  $D \cdot S$  values.

15. The method of claim 14, wherein at the end of not-coherent integration period several max values among  $S \cdot D$  not-coherent metrics for each Doppler frequency are separately selected, the selected values being read by a CPU, the saved values include not-coherent metrics, coherent metrics, and delay number.

16. The method of claim 10, further comprising: moving  $N$  cyclically shift registers forward at a rate of the FSNCO, wherein the input to a control accumulator is set to zero based on a configuration.

17. The method of claim 10, wherein addition of estimates during data sorting comprises the results of Doppler metrics during the period  $S \cdot k$  and Doppler metrics for each offset are separately sorted.

---